

#This code will be used to demonstrate lemma 6.11. We will show that $\tilde{B} = 0$ admits a single connected component in the region of the R^2 parameters.

$$A := -(w+z)^3 ((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4) :$$

$$B := x^2 y^5 w^2 z^2 (w+z)^2 (w-z) - x^2 y^2 (w+z)^3 (w^2 + z^2) w^2 z^2 (w-z) + w^4 y^3 z^4 (w+z)^3 + w^4 z^4 x^3 y^3 :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(collect(numer(expand(subs(parametrization, B))), a));

$$\begin{aligned} & -4 (-b+a)^3 (ab+1)^3 (b+a)^3 (b^2+1)^2 \left(\left(-\frac{1}{2} b^5 + b^7 \right) a^{10} + \left(\frac{3}{2} b^7 - 3 b^5 + b^3 \right) a^8 \right. \\ & + \frac{b^2 (b^2-5) (b^2+1)^3 a^7}{2} + \left(-\frac{1}{2} b - \frac{7}{2} b^5 + 3 b^3 \right) a^6 + \left(\frac{1}{2} - \frac{3}{2} b^8 - 4 b^6 \right. \\ & \left. - 3 b^4 \right) a^5 + \left(\frac{5}{2} b^3 - b - \frac{1}{2} b^7 \right) a^4 - (b^2+1)^3 a^3 + \left(-\frac{1}{2} b + b^5 \right) a^2 - \frac{b^3}{2} \Big) (a^2 \\ & + 1)^2 \end{aligned} \quad (1)$$

#Define $\tilde{B}$

$$\begin{aligned} \tilde{B} := & \left(\left(-\frac{1}{2} b^5 + b^7 \right) a^{10} + \left(\frac{3}{2} b^7 - 3 b^5 + b^3 \right) a^8 + \frac{b^2 (b^2-5) (b^2+1)^3 a^7}{2} + \left(-\frac{1}{2} b \right. \right. \\ & \left. - \frac{7}{2} b^5 + 3 b^3 \right) a^6 + \left(\frac{1}{2} - \frac{3}{2} b^8 - 4 b^6 - 3 b^4 \right) a^5 + \left(\frac{5}{2} b^3 - b - \frac{1}{2} b^7 \right) a^4 - (b^2 \\ & + 1)^3 a^3 + \left(-\frac{1}{2} b + b^5 \right) a^2 - \frac{b^3}{2} \Big) : \end{aligned}$$

#Calculating the intersection of $\tilde{B}$ with the line $b=a$

simplify(eval($\tilde{B}$, $a=b$))

$$\frac{b^3 (3 b^6 - 12 b^4 + 9 b^2 - 4) (b^2 + 1)^4}{2} \quad (2)$$

#Define b0

$b0 := \text{solve}(3 b^6 - 12 b^4 + 9 b^2 - 4 = 0, b, \text{useassumptions})$ assuming $b > 0$;

$$b0 := \frac{\sqrt{3 \cdot 7^2 / 3 + 3 \cdot 7^1 / 3 + 12}}{3}$$

(3)

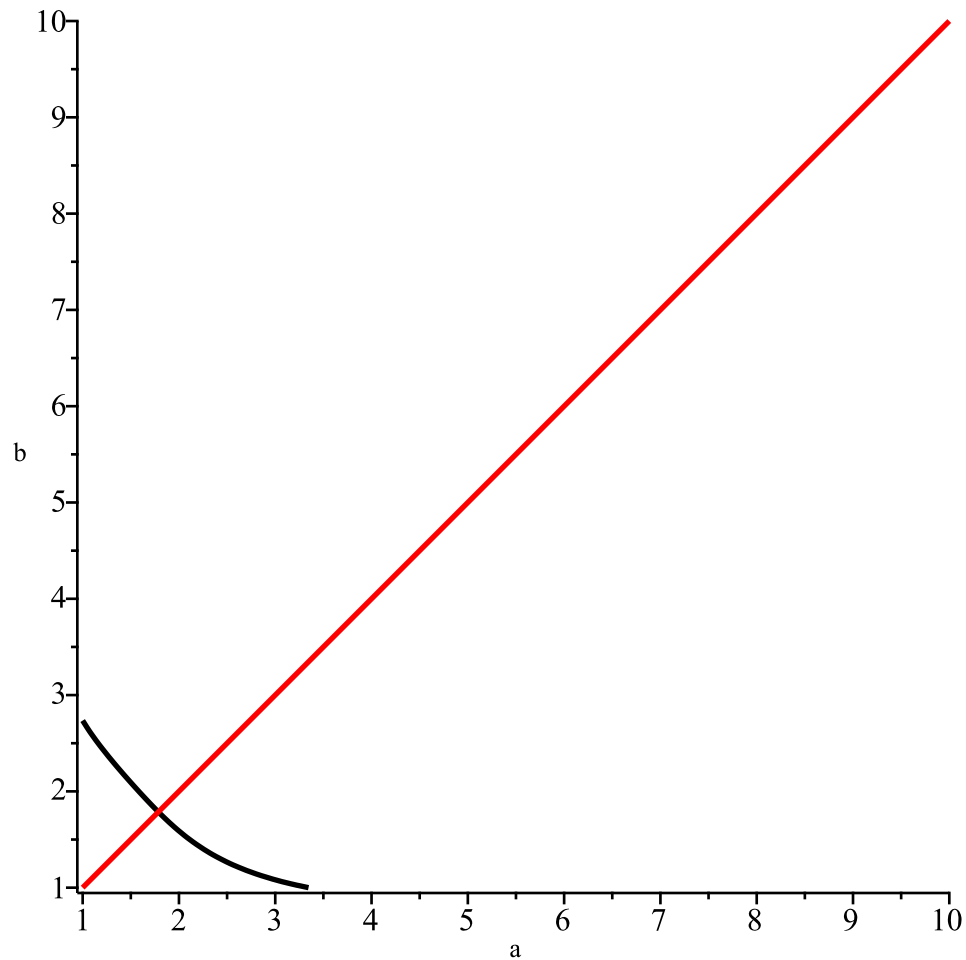
#Plot the curves $\tilde{n}=0$ and $\tilde{B}=0$ in R2.

$r := a - b :$

with(plots) :

implicitplot([$\tilde{B}=0$, $r=0$], a = 1 .. 10, b = 1 .. 10,
scaling = constrained,
gridrefine = 2,
thickness = 2,
color = [black, red],
labels = ["a", "b"],
title = "Graphic for $\tilde{B}=0$ in R2");

Graphic for $\tilde{B}=0$ in R2



#Apply the Möbius transformation on the rectangle $A1 = [1, 4) \times [1, b0)$ constructing B

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{1 + 4 \cdot c}{1 + c}, \\ b = \frac{1 + b0 \cdot d}{1 + d} \end{array} \right\} :$$

$B := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, \tilde{B}))), d) :$

#Compute the degree of B in c

$\text{degree}(B, c);$

10

(4)

#Define the coefficients $B_i = \text{coeff}(B, c, i)$

$\text{seq}(\text{evalf}(\text{assign}(\text{cat}('B', i), \text{coeff}(B, c, i))), i = 0 .. 10);$

#Compute the all degree of B_i in d

$\text{seq}(\text{degree}(\text{eval}(\text{cat}('B', i))), d), i = 0 .. 10);$

10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10

(5)

#From here we will analyze the signal variation of all coefficients B_i

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B10, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(6)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B9, d, i))), i = 10 .. 0, -1);$

$\text{fsolve}(B9 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1

0.03065234180

(7)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B8, d, i))), i = 10 .. 0, -1);$

$\text{fsolve}(B8 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1

0.3138706913

(8)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B7, d, i))), i = 10 .. 0, -1);$

$\text{fsolve}(B7 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1

0.9917799276

(9)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B6, d, i))), i = 10 .. 0, -1);$

$\text{fsolve}(B6 = 0, d, 0 .. \text{infinity});$

1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1

3.746873653

(10)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B5, d, i))), i = 10 .. 0, -1);$

$$fsolve(B5=0, d, 0 \dots \infty);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad (11)$$

$$seq(sign(evalf(coeff(B4, d, i))), i = 10 \dots 0, -1);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad (12)$$

$$seq(sign(evalf(coeff(B3, d, i))), i = 10 \dots 0, -1);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad (13)$$

$$seq(sign(evalf(coeff(B2, d, i))), i = 10 \dots 0, -1);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad (14)$$

$$seq(sign(evalf(coeff(B1, d, i))), i = 10 \dots 0, -1);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad (15)$$

$$seq(sign(evalf(coeff(B0, d, i))), i = 10 \dots 0, -1);$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad (16)$$

restart

#Define $\tilde{B}$

$$\tilde{B} := \left(\left(-\frac{1}{2} b^5 + b^7 \right) a^{10} + \left(\frac{3}{2} b^7 - 3 b^5 + b^3 \right) a^8 + \frac{b^2 (b^2 - 5) (b^2 + 1)^3 a^7}{2} + \left(-\frac{1}{2} b - \frac{7}{2} b^5 + 3 b^3 \right) a^6 + \left(\frac{1}{2} - \frac{3}{2} b^8 - 4 b^6 - 3 b^4 \right) a^5 + \left(\frac{5}{2} b^3 - b - \frac{1}{2} b^7 \right) a^4 - (b^2 + 1)^3 a^3 + \left(-\frac{1}{2} b + b^5 \right) a^2 - \frac{b^3}{2} \right) :$$

#Apply the Möbius transformation on the rectangle $A2 = [4, +\infty) \times [1, \infty)$ constructing B

$moebius := \{ a = 4 + c, \\ b = 1 + d \} :$

$B := \text{collect}(\text{numer}(\text{expand}(\text{subs}(moebius, \tilde{B}))), d) :$

#Compute the degree of B in c

$\text{degree}(B, c);$

10

(17)

#Define the coefficients $B_i = \text{coeff}(B, c, i)$
 $\text{seq}(\text{evalf}(\text{assign}(\text{cat}('B', i), \text{coeff}(B, c, i))), i = 0 .. 10);$

#Compute the all degree of B_i in d
 $\text{seq}(\text{degree}(\text{eval}(\text{cat}('B', i))), d), i = 0 .. 10);$

10, 10, 10, 10, 10, 10, 10, 10, 10, 7, 7, 7

(18)

#From here we will analyze the signal variation of all coefficients B_i

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B10, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(19)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B9, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(20)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B8, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(21)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B7, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(22)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B6, d, i))), i = 10 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(23)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B5, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(24)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B4, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(25)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B3, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(26)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B2, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(27)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B1, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(28)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B0, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(29)

restart

#Define $\tilde{B}$

$$\tilde{B} := \left(\left(-\frac{1}{2} b^5 + b^7 \right) a^{10} + \left(\frac{3}{2} b^7 - 3 b^5 + b^3 \right) a^8 + \frac{b^2 (b^2 - 5) (b^2 + 1)^3 a^7}{2} + \left(-\frac{1}{2} b \right. \right. \\ \left. \left. - \frac{7}{2} b^5 + 3 b^3 \right) a^6 + \left(\frac{1}{2} - \frac{3}{2} b^8 - 4 b^6 - 3 b^4 \right) a^5 + \left(\frac{5}{2} b^3 - b - \frac{1}{2} b^7 \right) a^4 - (b^2 \right. \\ \left. + 1)^3 a^3 + \left(-\frac{1}{2} b + b^5 \right) a^2 - \frac{b^3}{2} \right) :$$

#Define $b0$

$$b0 := \text{solve}(3 b^6 - 12 b^4 + 9 b^2 - 4 = 0, b, \text{useassumptions}) \text{ assuming } b > 0 :$$

#Apply the Möbius transformation on the rectangle $A3 = [b0, 4) \times [b0, 4)$ constructing B

$$\text{moebius} := \left\{ a = \frac{b0 + 4 \cdot c}{1 + c}, \right. \\ \left. b = \frac{b0 + 4 \cdot d}{1 + d} \right\} :$$

$$B := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, \tilde{B}))), d) :$$

#Compute the degree of B in c

$$\text{degree}(B, c);$$

10

(30)

#Define the coefficients $B_i = \text{coeff}(B, c, i)$

$$\text{seq}(\text{evalf}(\text{assign}(\text{cat}('B', i), \text{coeff}(B, c, i))), i = 0 .. 10);$$

#Compute the all degree of B_i in d

$$\text{seq}(\text{degree}(\text{eval}(\text{cat}('B', i))), d), i = 0 .. 10);$$

10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10

(31)

#From here we will analyze the signal variation of all coefficients B_i

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B10, d, i))), i = 10 .. 0, -1);$$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(32)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B9, d, i))), i = 10 .. 0, -1);$$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(33)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B8, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(34)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B7, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(35)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B6, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(36)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B5, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(37)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B4, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(38)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B3, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(39)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B2, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(40)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B1, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(41)

$$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(B0, d, i))), i = 10 \dots 0, -1);$$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$
(42)

#Check where $\tilde{B}$ is negative.

sturm(sturmseq(eval($\tilde{B}$, $b = 1$), a) , a , 3, 4);

1

(43)

#So (2,1) is to the left of $\tilde{B}(a0, 1)$

eval(eval($\tilde{B}$, $b = 1$), $a = 2$)

$$-\frac{4061}{2}$$

(44)